

Ideally, you are looking for investments that have consistent negative correlation. However you will quickly find with rolling correlation analysis that those investments do not exist. The next best option is to find investments that have a low correlation most of the time or a negative correlation some of the time.

As you may recall from [Chapter 3](#), correlation is measured using a numerical range between +1 and -1. When two investments move higher or lower than their average returns at the same time, they have *positive* correlation. When two investments move in opposite directions from their averages, they have a *negative* correlation. Two investments are considered noncorrelated or have 0 correlation when they do not move in any particular uniformity around their averages.

Correlations shift over time in unpredictable ways. The extent of the shift tends to depend on how different the underlying risks are in two investments types. Investments that are positively correlated in one period can become noncorrelated or negatively correlated in the next. Consequently, any single period correlation measurement between two investments is of limited use.

Unfortunately, most college finance books cite only a long-term average correlation number to explain the relationship between two asset classes. A simple long-term average is rather useless because it give readers the false impression that the correlation does not change. To make matters worse, the single correlation number is often used by financial advisors and fed into computer models to create “optimal” asset allocation for clients. This approach may tell advisors how their clients *should* have allocated their assets in the past, but it tells you nothing about the optimal allocation for the future because the correlations have already changed.

In this book, “rolling” correlations are used as a reference for study rather than a static long-term number. Rolling correlations allow you to see how often correlations between asset classes shift and by how much. You’ll get a general idea of the times when an investment helps a portfolio allocation because it had low or negative correlation, and the times it was of less help because the correlations were high.

[Figure 5-3](#) illustrates the changes in correlation between U.S. stocks and bonds that has occurred over rolling five-year periods. It’s a fine example of how unpredictable the correlations between two asset classes can be.